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Examining Consumer Impulsive Purchase Intention in Virtual AI Streaming: A S-O-R Perspective

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Abstract

Virtual AI-driven streamers have been gradually used in live commerce, and they may affect consumer impulsive purchase intention. Drawing on the stimulus–organism–response (S-O-R) model, this research examined consumer impulsive purchase intention in virtual AI streaming. Based on survey data from 411 predominantly young and educated virtual AI streaming users recruited through snowball sampling, we found that perceived responsiveness, perceived likeability, perceived expertise, and perceived anthropomorphism of virtual AI streamers are associated with trust and flow experience, both of which predict consumers' impulsive purchase intentions. The fsQCA identified two paths that lead to impulsive purchase intention. The results imply that live streaming platforms need to engender consumers' trust and flow experience in order to increase their impulsive purchase intention.

Keywords: virtual AI streaming; impulsive purchase; S-O-R

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1. Introduction

Live commerce, as an emerging e-commerce model, has quickly become a vital bridge connecting consumers with brands [1]. Compared with traditional online commerce, live commerce can provide consumers with a brand-new shopping experience through real-time interaction and visual display. The live commerce industry is developing rapidly, with its global market size reaching approximately US$918.9 million in 2023, and is expected to grow rapidly at a compound annual growth rate of 21.2% from 2024 to 2030 [2]. According to a CNNIC [3] report, the number of live streaming users in China reached 833 million, accounting for 75.2% of its internet user population, showing the great market potential of live commerce.

With the rapid development of artificial intelligence (AI) technology, the emergence of virtual AI streamers has further revolutionized the field of live commerce [4]. They bring unprecedented vitality and innovation to live commerce with their unique interactive methods and personalized features. Virtual AI streamers use advanced computer graphics, machine learning, and natural language processing technology to create a virtual image with both human appearance and intelligent interactive capabilities [5]. These virtual streamers can not only provide 24-h live broadcast services but also respond quickly to audience questions. Additionally, the training on vast datasets enables them to make professional product recommendations. Due to the limitations of real-person live streamers, such as limited energy, high costs, control over content, and potential personal scandals, more businesses are turning to virtual AI streamers [6]. Unlike traditional virtual streamers (like VTubers), which require real humans for motion capture and voice acting, virtual

AI streamers primarily rely on artificial intelligence for voice synthesis and behavior simulation, which are completely out of the control of real people behind the scenes. High-quality real-time rendering technology makes virtual AI streamers more realistic, while advancements in AI enhance their ability to interact effectively with audiences. Lu do Magalu, a virtual influencer from Brazil, has received a lot of attention and has nearly 30 million followers for promoting and marketing products of various brands on social platforms [7]. In April 2024, JD.com, a reputable Chinese e-commerce platform, held a live streaming with a virtual AI streamer modeled after its CEO Liu Qiangdong [8]. The one-hour event attracted 20 million viewers and generated over 50 million RMB Yuan in total sales.

Previous research has focused on the effect of real-person streamer on consumer impulsive purchase [9–11], and has seldom examined the effect of virtual AI streamer, which is an emerging application, on impulsive purchase. In other words, the effect mechanism of virtual AI streamer on consumer impulsive purchase intention remains a question. This research tries to fill the gap. Based on the stimulus–organism–response (S-O-R) model, we propose that the features of virtual AI streamers (perceived responsiveness, perceived likeability, perceived expertise, and perceived anthropomorphism) are associated with consumers' trust and flow experience, both of which further predict their impulsive purchase intention. The S-O-R model has been widely used to examine consumer behavior in live commerce [12–14]. It provides a useful lens to uncover the mechanisms underlying consumer behavioral decisions. Thus, it is appropriate to adopt the S-O-R model to explore consumer impulsive purchase intention in the virtual AI streaming. The results will disclose the effect mechanism of virtual AI streamer on consumer impulsive purchase intention and extend extant research that has focused on traditional live streaming but has seldom examined the emerging AI streaming. They may also provide insights for live streaming platforms to leverage AI streamers to engender consumer trust and flow experience, and further facilitate the impulsive purchase intention.

2. Literature Review

2.1. Virtual AI Streamer

Virtual streamer, also known as digital human or digital avatar, is a virtual character with an anthropomorphic appearance, which is controlled by humans or software [5]. Anthropomorphism means endowing virtual streamers with human attributes such as appearances, voices, and emotions [15]. Examples of virtual streamers include Luo Tianyi, Hatsune Miku, and Lil Miquela. Virtual streamers have been widely used in entertainment, commercial marketing, and live commerce. Miao et al. [16] divided digital humans or digital avatars into four types: simple avatars, superficial avatars, intelligent avatars, unrealistic avatars, and digital avatars. Zhang et al. [6] noted that digital avatars with the highest intelligence and highest realism will be the trend of the future. Early virtual streamers had a low level of intelligence and often had difficulty coping with unconventional situations during live broadcasts. Their performance was not impressive, and the overall live streaming experience was poor [17]. In contrast, AI-powered virtual streamers have human-like appearances, high intelligence, and almost the same live broadcast capability as real people [18]. Based on large language models, virtual AI streamers have more intelligence and can autonomously interact with consumers. Using the inference ability, they can better understand consumer needs and offer the relevant information to stimulate consumers' impulsive purchase intention [19].

Previous studies have compared the differences between real-person streamers and virtual AI streamers, and found that real-person streamers trigger higher purchase intentions than virtual AI streamers due to the high intimacy and quick responses [4]. However,

virtual AI streamers have lower costs and can maintain a stable working state without interruption, while real-person streamers may be fatigued or have mood swings, which will affect the quality of live streaming [20]. The anthropomorphic characteristics of virtual AI streamers can have a positive impact on consumers' intention to accept virtual live streaming [21]. In addition, innovation resistance, shopping motivation, and personality can also influence consumers' switch from traditional live streaming to virtual AI streaming [22].

As evidenced by these studies, they have compared the differences between real-person streamers and virtual AI streamers and examined consumer switching from traditional live streaming to virtual AI streaming. Little research has examined consumer impulsive purchases in the context of virtual AI streaming. This research will draw on the S-O-R to explore the effect of virtual AI streamers on consumer impulsive purchase intention.

2.2. Live Commerce and Impulsive Purchase

Live commerce offers consumers a new way to shop. Compared to one-way communication methods such as pictures, texts, or videos, live commerce combines real-time social interaction with e-commerce and displays products in a very intuitive way, which can help consumers better understand the products and decide whether to buy them, making it significantly superior to traditional shopping methods. Live commerce emerged in 2016 and developed rapidly during the COVID-19 pandemic. As users stay at home, their demand for online shopping has increased, which makes live shopping even more popular. A large body of literature has explored consumers' purchasing intentions in live commerce. Among them, the effect of live streamer characteristics on consumer purchase behavior has attracted much attention, such as language style [23], visual appeal [24], quick response to unexpected events [25], and the relationship with consumers [26]. These studies show the dominant role of live streamers in live commerce.

Impulsive purchase refers to the unplanned, unthought, and sudden buying decisions, accompanied by a strong tendency to buy impulsively [27]. An impulsive purchase is usually caused by internal emotions or external environmental stimulation. Compared with traditional regular purchases, impulsive purchase often has a very short decision-making process. Consumers may make purchase decisions due to temporary pleasure, stress relief, or social influence [28]. Impulsive purchases originated in offline retail. With the development of e-commerce, research on impulsive purchases has been conducted in areas such as online commerce, social commerce, and live commerce. Existing studies have shown that online environments are more likely to induce consumers to make impulsive purchases than offline environments [29]. Much literature has studied the determinants of consumer impulsive purchase in live commerce, such as interactive factors in the live broadcast room [30], streamers' characteristics [11], social media influencers and live contents [9]. These studies offer the basis for this research to examine consumer impulsive purchases under the emerging mode of virtual AI streaming.

2.3. Stimulus–Organism–Response (S-O-R)

The S-O-R model, which was proposed by Mehrabian and Russell in 1974, explores how individuals' psychology and cognition (organism) change in response to external environmental stimuli (stimulus), which leads to behavioral responses. In live streaming, the interactions with virtual AI streamers may stimulate consumers' positive emotional experience and impulsive purchase intention. Thus, we propose that the features of virtual AI streamers (stimulus), which include perceived responsiveness, likeability, expertise, and anthropomorphism, are associated with consumer trust and flow experience (organism), both of which further lead to impulsive purchase intention (response). The S-O-R model has been widely used to study individual behavior decisions, including brand marketing,

online shopping, and social commerce. Wu and Li [31] argued that marketing factors in social commerce influence customer value, which in turn affects customer loyalty. Islam and Rahman [32] explored the impact of characteristics of online brand communities on user engagement and brand loyalty. Chen and Yao [33] stated that website quality and promotional activities in mobile auctions affect normative evaluations and positive emotions, which further affect users' impulsive purchases.

The S-O-R model has also been used to study consumer purchase behavior in live commerce. Guo et al. [34] found that the functionality of live streaming rooms affects consumers' perceived value and uncertainty, which in turn affects purchase intention. Lee and Chen [35] found that live streamer characteristics and product characteristics affect impulsive purchase intention through perceived pleasure and perceived usefulness. Cui et al. [36] noted that cognitive factors influence users' impulsive purchase intention through flow experience. Wu and Huang [37] stated that consumer trust mediates the effect of perceived value on consumers' intention to continue purchasing in live streaming. Liang et al. [38] argued that tourism live streaming features affect flow experience and trust, both of which in turn boost impulsive travel intentions. Wen et al. [39] reported that key opinion leader (KOL) characteristics such as professionalism and attractiveness enhance users' pleasure, arousal, and trust, which further affect impulse buying intentions.

In summary, there are two research gaps in the literature. First, extant research has focused on traditional live streaming and has seldom examined virtual AI streaming, which is an emerging mode. This research examined the effect of virtual AI streamer features on consumer impulsive purchase intention. The results enrich extant research on live streaming. Second, the mechanism underlying consumer impulsive purchase intention in virtual AI streaming remains a black box. Thus, this research adopts the S-O-R model as the theoretical framework to uncover the development process of consumer impulsive purchase intention in virtual AI streaming. The results advance our understanding of AI streaming consumer purchase decisions.

3. Research Model and Hypotheses

The research model is shown in Figure 1.

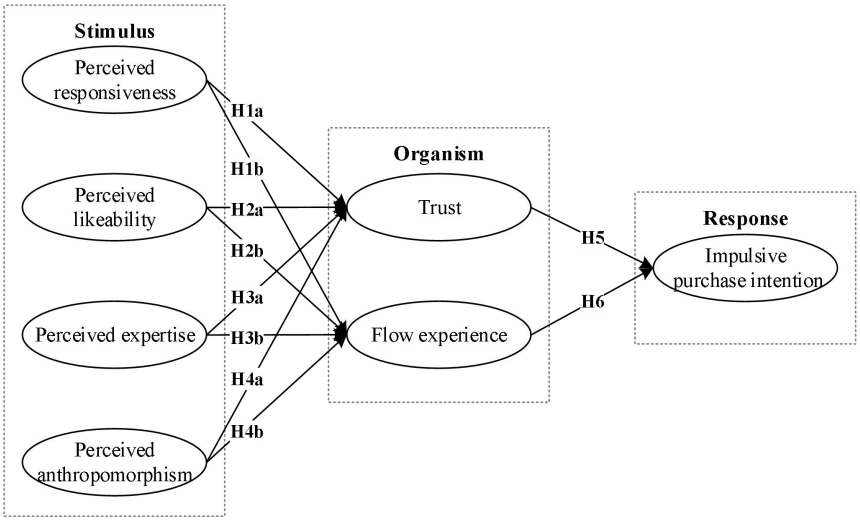


Figure 1. Research model.

3.1. Perceived Responsiveness

Perceived responsiveness reflects the ability of a virtual AI streamer to respond to the needs of consumers. When the AI streamer can respond quickly and accurately to users' real-time comments, including bullet screen comments, users will feel that their needs

are valued. This timely feedback can enhance users' trust in the virtual AI streamer, and they feel that the system is reliable and efficient. In the field of e-commerce, perceived responsiveness is a key indicator for evaluating the quality of an online store [40,41]. High responsiveness means that customer needs can be met in a timely manner, which in turn affects their satisfaction and purchase intention. This is especially important for live streaming that emphasizes real-time interactions. For example, Gao et al. [4] found that the response speed of e-commerce live streamers has a positive impact on consumers' purchasing intention.

Similarly, high responsiveness can enhance the immersion of live streaming consumers, making them feel that they are part of the live streaming system rather than passive viewers. This sense of immersion is an important part of the flow experience, enabling consumers to become more deeply engaged in live streaming activities. Noort et al. [42] found that the interactivity of brand websites has a significant impact on consumers' behavioral responses. Wang et al. [43] stated that the prompt response of the live streamer has a positive impact on the audience's flow experience. Prior studies also found that the responsiveness of the live streamer in online lectures live streaming [44] and tourism live streaming [45] can enhance the audience's flow experience. Similarly, responsiveness is also a key factor for AI chatbots to shape user experience [46]. The ability to respond quickly enables virtual AI streamers to strengthen their connections with users [47], enhance user engagement and satisfaction, and improve user experience. Therefore, we suggest the following:

H1a. *Perceived responsiveness is associated with consumer trust in the virtual AI streamer.*

H1b. *Perceived responsiveness is associated with consumer flow experience in virtual AI streaming.*

3.2. Perceived Likability

Perceived likability reflects a user's intuitive impressions and feelings toward a virtual AI streamer. When users have a high level of likeability toward an AI streamer, they are more likely to develop a psychological sense of closeness and acceptance. They may forget the virtuality of the virtual AI streamer to some extent, thereby improving overall trust. Previous research indicates that consumers tend to develop positive emotions toward live streamers they find likable, and may transfer these positive feelings to the products being promoted [48]. Users are more likely to trust likable streamers [49] and make impulsive purchases [11]. Previous research has found that perceived likeability affects swift guanxi between consumers and streamers [50,51]. Similarly, Zhou et al. [52] noted that the likeability of the animated speaking character affects parasocial interaction relationships. Nagel et al. [49] argued that suppliers' likeability affects purchasers' trust in suppliers. These studies show that perceived likability helps facilitate mutual relationships, which include trust.

When consumers watch virtual AI streaming, perceived likability makes them more attentive to the content and more willing to interact with AI streamers, which can enhance the flow experience to some extent. Additionally, a high likability toward virtual AI streamers directly affects users' enjoyment during interactions. Users are more focused and engaged while watching the live streaming, leading to a flow experience. Gao et al. [24] found that virtual streamers' likeability enhances social presence and telepresence. Zheng et al. [53] argued that streamer attractiveness (similar to likeability) affects viewers' flow experience. Similarly, Tang et al. [12] found that streamers' physical attractiveness affects consumers' flow experience. Yang and Lee [54] noted that visual attractiveness influences the flow experience of augmented reality users. Therefore, we propose the following:

H2a. *Perceived likability is associated with consumer trust in the virtual AI streamer.*

H2b. *Perceived likability is associated with consumer flow experience in virtual AI streaming.*

3.3. Perceived Expertise

Perceived expertise of an AI streamer reflects its knowledge and understanding of the products, as well as the accuracy of the information it conveys. Research shows that a streamer's product expertise helps build consumer trust [55]. When users believe that the live streamer is highly professional and credible, they will trust the information provided by the live streamer and make purchasing decisions accordingly. The expertise of a virtual AI streamer is directly related to user evaluation of its credibility. Previous studies have shown that the expertise of AI virtual assistants will significantly affect user trust [56]. When users feel that the virtual AI streamer has profound knowledge or professional skills, they are more likely to consider the AI streamer as a trustworthy entity and are willing to follow their opinions or suggestions.

In addition, the live streamer's professionalism and expertise also have a positive impact on the consumer flow experience. The virtual AI streamers, built on general AI and trained for specific shopping scenarios, have a deep understanding of the products or services they recommend. This allows them to output high-quality content, making the live streaming process more engaging. As a result, they capture users' attention, helping them focus on the live content and making it easier for them to achieve a flow experience. Prior research has identified the effect of perceived expertise on flow experience in mobile short video apps [57], social commerce [58], and live commerce [59,60]. Consistent with these studies, we suggest the following:

H3a. *Perceived expertise is associated with consumer trust in the virtual AI streamer.*

H3b. *Perceived expertise is associated with consumer flow experience in virtual AI streaming.*

3.4. Perceived Anthropomorphism

Perceived anthropomorphism refers to giving non-human entities human characteristics, such as appearance, expression, and action. Previous studies have shown that anthropomorphic features of intelligent personal assistants can enhance their task attractiveness (perception of their ability to complete tasks) and social attractiveness (intention to become friends with them), thereby improving cognitive and emotional trust [15]. Adding anthropomorphic features to AI systems can significantly increase user trust [61]. When users interact with systems or entities that appear "human-like", they may feel more comfortable and build trust. The impact of anthropomorphism on trust may also vary depending on the product type. Compared with hedonic products, the effect of anthropomorphism on trust in utilitarian products will be more significant [21]. Anthropomorphism makes the interaction between virtual AI streamers and users more personal and human-like. Users will view the anthropomorphic virtual AI streamer as a friendly partner rather than a cold machine, and they are more willing to interact with AI that exhibits anthropomorphic characteristics. This shortened emotional distance between both parties and improved user trust.

Meanwhile, existing studies have also shown that virtual agents with anthropomorphic characteristics can effectively enhance consumer flow experience [62] and improve user satisfaction [63]. When interacting with the highly anthropomorphic virtual AI streamer, users may feel as if a real shopping guide is introducing the products to them, which helps the live streamer establish a deep connection with the user. Anthropomorphized virtual AI streamers can facilitate emotional interactions and support, making it easier for consumers

to understand and accept the content. This enhances consumer engagement and focus and leads to a flow experience.

H4a. *Perceived anthropomorphism is associated with consumer trust in a virtual AI streamer.*

H4b. *Perceived anthropomorphism is associated with consumer flow experience in virtual AI streaming.*

3.5. Trust

Trust reflects a willingness to be vulnerable based on the positive expectation toward future outcomes [64]. It often includes three dimensions of ability, integrity, and benevolence [65]. Ability means that an individual has the knowledge and skills to fulfill his or her tasks. Integrity means that an individual will keep his or her promises and not deceive other people. Benevolence means that an individual is concerned with other people's interests. Trust plays a vital role in facilitating online transactions, and it is considered a key factor for reducing perceived risk and ensuring e-commerce success [66]. In e-commerce transactions, the lack of face-to-face interaction between buyers and sellers increases perceived uncertainty and risk. Trust can help mitigate perceived risk and facilitate the transaction [67]. Trust not only affects consumers' regular purchasing decisions but also affects their impulsive purchase behavior. The higher the consumer's trust in the streamer, the more likely they are to engender impulsive purchase intentions [11]. Similarly, when consumers perceive a virtual AI streamer as trustworthy, they are more likely to be attracted and develop impulsive purchase intentions. Consumers believe that the virtual AI streamer has sufficient ability and integrity to provide quality information and services. This makes users more inclined to adopt the streamer's recommendations, and reduces concerns in purchasing decisions, which further triggers impulsive purchase intentions. Thus, we suggest the following:

H5. *Trust is associated with consumer impulsive purchase intention.*

3.6. Flow Experience

Flow experience reflects a state in which an individual is fully focused on an activity [68]. Flow experience is characterized by immersion in the current activity, and it has a positive impact on consumer behavior [69,70]. In the field of e-commerce, there is a significant correlation between flow experience and user behaviors such as continuation intention, purchase intention, and impulsive purchase [69]. Flow experience can enhance consumers' shopping experience, making them focused on current purchasing activities, thereby increasing the possibility of impulsive purchases. Previous studies have found that the flow experience of users in live streaming significantly affects their attitudes [42], consumption intention [43], and impulsive purchase intention [36]. Flow experience is usually accompanied by a high degree of pleasure and satisfaction, and it also leads to higher attention and participation. Therefore, when users watch virtual AI streaming and immerse themselves in it, they are more likely to make immediate and irrational purchase decisions, that is, impulsive purchases. Thus, we propose,

H6. *Flow experience is associated with consumer impulsive purchase intention.*

4. Methodology

4.1. Instrument Development

The research model includes seven constructs. The measurement items were adapted from the existing literature to ensure content validity. Whenever possible, these items were

derived from the literature on AI or live streaming, which are consistent with the context of this research. When the instrument was developed, it was tested among ten users who had rich experience using virtual AI streaming. Based on their comments, we revised a few items to improve clarity and understandability. Table 1 lists the measurement items and their sources. All items were measured using a five-point Likert scale.

Table 1. Constructs and measurement items.

Constructs	Items	Items Content	Source
Perceived responsiveness (PR)	PR1	This virtual AI streamer is very happy to communicate with me.	[24]
	PR2	This virtual AI streamer can answer my questions and requests in a timely manner.	
	PR3	The reply of this virtual AI streamer is closely related to my problems and requests.	
Perceived likeability (PL)	PL1	This virtual AI streamer is likable.	[24,71]
	PL2	This virtual AI streamer is friendly.	
	PL3	This virtual AI streamer is pleasant.	
	PL4	This virtual AI streamer is nice.	
Perceived expertise (PE)	PE1	This virtual AI streamer is an expert.	[10]
	PE2	This virtual AI streamer is experienced.	
	PE3	This virtual AI streamer is knowledgeable.	
Perceived anthropomorphism (PA)	PA1	This virtual AI streamer acts in a natural way.	[72]
	PA2	This virtual AI streamer is more human-like.	
	PA3	This virtual AI streamer appears to be conscious.	
	PA4	This virtual AI streamer seems to be alive rather than artificial.	
Trust (TRU)	TRU1	I believe the information released by the virtual AI streamer.	[10,73]
	TRU2	I believe that the virtual AI streamer will not deliberately release false information to deceive consumers.	
	TRU3	I believe that the products recommended by the virtual AI streamer are of high quality.	
Flow experience (FE)	FE1	When watching the virtual AI streaming, I felt that time passed very quickly.	[74]
	FE2	I was curious when watching the virtual AI streaming.	
	FE3	I was very focused when watching the virtual AI streaming.	
Impulsive purchase intention (IPI)	IPI1	After watching the virtual AI streaming, I will buy some products that I had not planned to buy before.	[75]
	IPI2	After watching the virtual AI streaming, I found myself wanting to buy some things I had not planned to buy.	
	IPI3	After watching the virtual AI streaming, I will buy some products without thinking.	

4.2. Data Collection and Sample Demographics

This research developed an online questionnaire to collect data. Users who had experience using virtual AI streamers were invited to participate in the survey. They were asked to fill out the questionnaire based on the experience of their favorite virtual AI streamer. Snowball sampling was adopted in this research. The respondents were encouraged to forward the questionnaire to their friends in order to expedite data collection. We scrutinized all responses and removed a few invalid ones, such as those that had no

experience using virtual AI streamers and those that had the same answer for all questions. As a result, we obtained 411 valid responses.

Among them, 48.42% were male and 51.58% were female, showing a gender balance. 71.29% of the respondents were below 40 years old, and 63.5% held a bachelor's or higher degree. This suggests that young people with high levels of education are the main group of virtual AI streaming users. 76.64% of the respondents watched virtual AI streaming at least once a week, and 64.96% of the respondents watched it for more than 30 min each time. The results show that 63.75% of the respondents have watched IP-based digital humans (such as virtual idols Hatsune Miku and Luo Tianyi), 56.69% have watched functional AI streamers (such as virtual AI streamer Liu Qiangdong and Lil Miquela), and 39.9% have watched avatar-based digital humans (such as CCTV "Little Sa Beining" and Neuro-sama).

4.3. Data Analysis Methods

In this research, we adopted a mixed method of structural equation modeling (SEM) and fuzzy-set qualitative comparative analysis (fsQCA) to conduct data analysis. During SEM, we first examined the measurement model to test reliability and validity. Then, we examined the structural model to test research hypotheses and model fitness. However, SEM focuses on the "net effect" of independent variables on the dependent variable, and neglects the complex causal relationship formed by the interdependence of multiple antecedent variables [76]. From a holistic perspective, fsQCA regards the research object as a configuration of the different combinations of antecedent variables [77], and it can identify the conditional combination that triggers the outcome variable. Therefore, this research uses fsQCA to examine the different configurations that lead to impulsive purchase intention.

To test common method variance (CMV), we first performed Harman's single-factor test. The results indicated that the largest variance explained by an individual factor is 12.36%. Thus, none of the factors can explain the majority of the variance. Second, we modeled all items as the indicators of a factor representing the method and re-estimated the model. The results indicated a poor fit. For example, the goodness of fit index (GFI) is 0.704 (<0.90), and the root mean square error of approximation (RMSEA) is 0.136 (>0.08). The results of both tests indicated that CMV is not a significant problem in this research.

5. Results

5.1. Structural Equation Modeling (SEM)

5.1.1. Measurement Model

First, we used SPSS 26 to test the reliability of the scale. As listed in Table 2, the Alpha coefficient of each variable ranged from 0.797 to 0.858, showing good internal consistency. The composite reliability (CR) values were between 0.798 and 0.859, factor loadings were greater than 0.70, and the average variance extracted (AVE) values were above the threshold of 0.50, indicating that the scale had good convergent validity. The variance inflation factor (VIF) values are below 3, showing that multicollinearity does not exist in this research. In addition, as shown in Table 3, the square root of AVE (shown at the diagonal) is greater than the correlation coefficient between variables, showing good discriminant validity. Table 4 lists the cross-loading matrix. Each item has higher loading on its corresponding factor than the cross-loading on other factors, indicating good validity.

Table 2. Reliability and validity.

Constructs	Items	Loading	CR	AVE	Alpha	VIF
Perceived responsiveness (PR)	PR1	0.780	0.807	0.582	0.806	1.393
	PR2	0.765				
	PR3	0.743				
Perceived likability (PL)	PL1	0.752	0.831	0.552	0.832	1.456
	PL2	0.722				
	PL3	0.769				
	PL4	0.729				
Perceived expertise (PE)	PE1	0.720	0.807	0.582	0.805	1.691
	PE2	0.752				
	PE3	0.814				
Perceived anthropomorphism (PA)	PA1	0.767	0.859	0.604	0.858	1.723
	PA2	0.771				
	PA3	0.814				
	PA4	0.756				
Trust (TRU)	TRU1	0.771	0.798	0.569	0.797	1.729
	TRU2	0.719				
	TRU3	0.772				
Flow experience (FE)	FE1	0.803	0.834	0.625	0.833	1.512
	FE2	0.799				
	FE3	0.770				
Impulsive purchase intention (IPI)	IPI1	0.796	0.832	0.623	0.832	-
	IPI2	0.785				
	IPI3	0.787				

Table 3. Correlation matrix.

Constructs	PR	PL	PE	PA	TRU	FE	IPI
PR	0.763						
PL	0.501	0.743					
PE	0.474	0.497	0.763				
PA	0.430	0.464	0.641	0.777			
TRU	0.530	0.543	0.652	0.639	0.754		
FE	0.474	0.507	0.528	0.558	0.521	0.791	
IPI	0.553	0.575	0.600	0.605	0.656	0.596	0.789

Table 4. Cross-loading matrix.

	PA	PL	FE	PR	PE	IPI	TRU
PR1	0.098	0.165	0.139	0.817	0.054	0.132	0.101
PR2	0.144	0.100	0.157	0.772	0.135	0.124	0.149
PR3	0.079	0.175	0.064	0.779	0.121	0.144	0.134
PL1	0.114	0.745	0.136	0.163	0.174	0.090	0.115
PL2	0.122	0.758	0.123	0.097	0.057	0.124	0.163
PL3	0.088	0.741	0.111	0.194	0.208	0.215	0.019
PL4	0.128	0.793	0.117	0.065	0.000	0.098	0.159
PE1	0.183	0.107	0.194	0.159	0.794	0.055	0.098
PE2	0.244	0.168	0.103	0.002	0.634	0.190	0.324
PE3	0.199	0.143	0.111	0.166	0.776	0.199	0.171
PA1	0.780	0.172	0.147	0.064	0.060	0.138	0.203
PA2	0.743	0.077	0.173	0.095	0.289	0.134	0.088

Table 4. *Cont.*

	PA	PL	FE	PR	PE	IPI	TRU
PA3	0.807	0.105	0.139	0.109	0.075	0.151	0.242
PA4	0.698	0.154	0.136	0.150	0.297	0.161	0.080
TRU1	0.208	0.172	0.151	0.146	0.155	0.189	0.732
TRU2	0.108	0.140	0.076	0.202	0.276	0.164	0.689
TRU3	0.257	0.165	0.147	0.118	0.095	0.150	0.775
FE1	0.149	0.146	0.798	0.104	0.138	0.197	0.113
FE2	0.180	0.143	0.789	0.130	0.191	0.159	0.069
FE3	0.184	0.180	0.782	0.152	0.064	0.102	0.170
IPI1	0.250	0.156	0.193	0.191	0.132	0.736	0.147
IPI2	0.136	0.183	0.197	0.145	0.205	0.741	0.186
IPI3	0.182	0.204	0.128	0.154	0.096	0.785	0.192

5.1.2. Structural Model

Next, we used AMOS 26 to analyze the structural model and estimate path coefficients. The results are shown in Figure 2. All hypotheses were supported. The explained variance (R^2) of trust, flow experience, and impulsive purchase intention is 62%, 45%, and 58%, respectively. Table 5 presents the model fit results. All fit indices have better actual values than the recommended values, indicating that the model has a good fit.

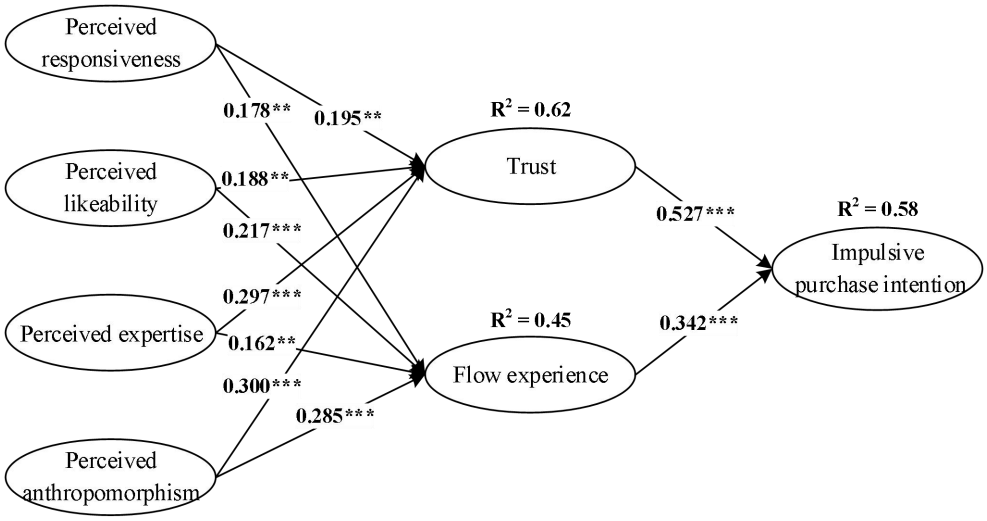


Figure 2. Path coefficients and significance. (Note: **, $p < 0.01$; ***, $p < 0.001$)

Table 5. Model fit indices.

Fit Index	χ^2/df	GFI	AGFI	CFI	NFI	RMSEA
Recommended value	<3	>0.9	>0.8	>0.9	>0.9	<0.08
Actual value	1.768	0.915	0.890	0.964	0.921	0.043

5.2. Fuzzy-Set Qualitative Comparative Analysis (fsQCA)

5.2.1. Model Construction

We selected six antecedent variables: perceived responsiveness, perceived likeability, perceived expertise, perceived anthropomorphism, trust, and flow experience. The items of these variables were averaged and calibrated using the fsQCA 4.0 software, following the standards of 5%, 95%, and the 50% crossover point proposed by Ragin [78]. The necessity analysis showed that the consistency of each antecedent variable was below 0.9, showing

that no single antecedent variable is a necessary condition for the outcome variable. Thus, the conditional configuration analysis can be performed.

5.2.2. Configurational Analysis

Based on the number of antecedent variables, a 64-row (2^6) truth table was constructed, with each row representing a possible combination of antecedent variables. Following Pappas and Woodside [77], we set the frequency threshold at 8, the consistency threshold at 0.8, and the PRI threshold at 0.75. The results are shown in Table 6.

Table 6. Configuration paths of impulsive purchase intention.

Conditional Variables	Impulsive Purchase Intention	
	Path 1	**Path 2**
Perceived responsiveness	•	
Perceived likeability	•	•
Perceived expertise	•	•
Perceived anthropomorphism		•
Trust	●	●
Flow experience	●	●
Raw coverage	0.443	0.457
Unique coverage	0.027	0.041
Consistency	0.948	0.934
Overall solution coverage	0.484	
Overall solution consistency	0.928	

(Note: "●"and "•" indicate the presence of a core condition and a peripheral condition, respectively. The "blank" indicates that the condition is optional.)

6. Discussion

As Figure 2 shows, all hypotheses are supported. The results indicated that four features of virtual AI streamer (stimulus), which include perceived responsiveness, likeability, expertise, and anthropomorphism, are associated with trust and flow experience (organism), both of which further predict impulsive purchase intention (response). Thus, the S-O-R offers a useful framework to uncover the mechanism underlying consumer impulsive purchase intention in virtual AI streaming.

The results indicated that perceived responsiveness of virtual AI streamer has a significant impact on both consumer trust (H1a) and flow experience (H1b), which is consistent with previous research [24,79]. Perceived responsiveness is an important factor to evaluate whether the virtual AI streamer has the ability to understand and meet consumer needs. When the virtual AI streamer can respond to consumers' questions or needs quickly and accurately, they tend to believe that the streamer has certain intelligence and capabilities, thereby enhancing their trust. In addition, the virtual AI streamer's quick responses can provide users with a smooth access experience, thereby promoting the generation of a flow experience.

Perceived likability is positively associated with trust (H2a) and flow experience (H2b). This is similar to previous research on virtual assistants [56,80]. Likeability is usually based on the virtual AI streamer's appearance, performance, and other aspects. If users feel that the virtual AI streamer is friendly, kind, or reliable, they are more likely to trust the information and suggestions provided by the virtual AI streamer. Additionally, when consumers develop a favorable impression of a virtual AI streamer, they are more likely to be deeply engaged in the interactions, leading to an immersive experience. A high level of likeability makes consumers enjoy the interaction process, and this positive emotion facilitates their flow experience.

The study found that perceived expertise of virtual AI streamer has significant effects on trust (H3a) and flow experience (H3b). The more professional the streamers, the stronger the consumers' sense of trust. This is in line with previous research [11]. When a virtual AI streamer demonstrates a high level of knowledge and skills, users are more likely to trust their opinions and recommendations because they perceive the streamer as competent and experienced. This sense of trust stems from the user's recognition of the streamer's professional abilities, which is a rational judgment. Moreover, a professional virtual AI streamer can deliver high-quality content, minimizing redundant information and errors. This reduces the cognitive load on users, allowing them to focus more on the experience without the need to filter out irrelevant information, thereby enhancing the overall user experience.

Perceived anthropomorphism of virtual AI streamer is also associated with both trust (H4a) and flow experience (H4b). The anthropomorphic features of virtual AI streamers—such as voice, facial expressions, and gestures—make it easier for users to form emotional connections with them. When consumers recognize and accept these anthropomorphic traits, they are more likely to perceive the AI as a "human-like" entity, which enhances their trust [81]. Sun et al. [82] also found that anthropomorphism affects customer engagement with virtual live shopping platforms. Additionally, this human-like interaction allows users to achieve a deeper sense of immersion, making it easier for them to enter a flow state during interactions with the virtual AI streamer.

Both trust (H5) and flow experience (H6) predict impulsive purchase intention. This is consistent with previous research on online shopping [69,83]. Compared to flow experience ($\beta = 0.342$), trust has a stronger effect ($\beta = 0.527$) on impulsive purchase intention, indicating that trust is the primary factor influencing impulsive purchase. This may be because trust is related to the results of consumers' purchase decisions, such as whether the purchased products or services are reliable and meet expectations, while the flow experience only reflects the process of consumer interaction with the virtual AI streamer. When consumers develop trust in the virtual AI streamer, they are more likely to accept the information and conduct an impulsive purchase. In addition, flow experience, as an emotional immersion, also influences consumer impulsive purchase decisions.

As listed in Table 5, the fsQCA indicated that there are two paths that trigger the impulsive purchase intention. In both paths, trust and flow experience are the common core conditions, suggesting that they are the crucial factors leading to consumer impulsive purchase intention. This result is consistent with the SEM results. In addition, perceived likeability and perceived expertise are the common peripheral conditions of both paths, indicating their indispensable role in engendering impulse purchase intention. It is worth noting that perceived responsiveness is a peripheral condition of Path 1, but it is optional in Path 2. In contrast, perceived anthropomorphism is a peripheral condition of Path 2, but it is optional in Path 1. This suggests that both paths are complementary to each other. While the former one highlights the utilitarian value (prompt responses) of the virtual AI streamer, the latter one highlights the emotional value (anthropomorphic experience). This result reveals the different development paths of consumer impulsive purchase intention in virtual AI streaming. For live streaming platforms, they need to provide prompt responses to those consumers who are concerned with the AI streamer utility, while providing human-like interactions to those consumers who are concerned with the emotional experience using the AI streamer.

7. Theoretical and Practical Implications

This research makes four contributions. First, existing research has primarily focused on traditional e-commerce live streaming and has seldom examined virtual AI streaming,

which is an emerging mode. This study examined the impact of virtual AI streamers on consumer impulsive purchase intention. The results enrich the research on live streaming. Second, based on the S-O-R, this research found that the features of virtual AI streamer are associated with trust and flow experience, both of which further predict impulsive purchase intention. The results disclose the formation mechanism of consumer impulsive purchase intention in the virtual AI streaming. Third, compared to flow experience, trust has a stronger impact on impulsive purchase intention, indicating that rational judgment rather than emotional experience is the primary factor affecting impulsive purchase intention. This extends our understanding of consumer impulsive purchase behavior. Fourth, the fsQCA indicated that there are two configuration paths triggering impulsive purchase intention. While a path focuses on the responsiveness, the other one focuses on the anthropomorphism. The results reveal the complexity of impulsive purchase intention formation in the virtual AI streaming.

The results have a few practical implications for live streaming platforms. First, they need to enhance the technical capabilities of virtual AI streamers, such as responsiveness and expertise. Virtual AI streamers need to optimize their algorithms and provide quick responses during the interaction with consumers. This may engender a fluid experience for consumers. In addition, AI streamers should not only possess general knowledge but also be trained with company or product-specific data. This will enable them to acquire more professional knowledge, respond to consumer needs accurately, and build consumer trust. Second, live streaming platforms also need to pay attention to the emotional characteristics of virtual AI streamers, such as likeability and anthropomorphism. Platforms can establish a virtual AI streamer persona that matches the corporate brand to improve consumer likability towards the virtual AI streamer. Additionally, the virtual AI streamer can provide emotional support and comfort through human-like interactions to enhance the consumer experience. These features may develop emotional connections between AI streamers and consumers and facilitate impulsive purchase intention.

8. Conclusions

Based on the S-O-R model, this research investigated consumer impulsive purchase intention in virtual AI streaming. We found that perceived responsiveness, perceived likeability, perceived expertise, and perceived anthropomorphism of virtual AI streamers are associated with trust and flow experience, both of which predict consumer impulsive purchase intentions. The results provide valuable insights for live streaming platforms to promote consumer impulsive purchase intention.

This research has a few limitations, which may offer potential directions for future research. First, we used the snowball sampling method to collect data. The results indicated that our sample is mainly composed of young people with high levels of education. Although they are the main group of virtual AI streaming users, future research may adopt probability sampling and generalize our results to other samples, such as middle-aged or elderly users. Second, this study mainly examined the impact of trust and flow experience on impulsive purchase intention. Although both factors are found to have strong effects on impulsive purchase intention, future research may examine the potential effect of other factors such as perceived value, satisfaction, and perceived risk. Third, this study mainly used cross-sectional data, which cannot accurately predict the causal relationship. Future research could collect longitudinal data to test the causal effects between variables and examine the development of impulsive purchase behavior over time. Fourth, this research is conducted in China, where AI and related applications are developing rapidly. As culture (such as guanxi) may affect consumer purchase decisions, future research may compare our results to those in Western cultures and identify the effect of cultural factors on virtual

AI streaming consumer behavior. Fifth, this research mainly adopted the S-O-R model as the theoretical base. Future research may adopt other theories, such as the technology acceptance model, the use and gratification theory, and the social presence theory, to enrich the understanding of consumer impulsive intention in virtual AI streaming.

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